

Normal Distributions

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: Conditional Probability

Let A and B be two events such that $P(A) > 0$ and $P(B) > 0$. Suppose that $P(A|B) = 1$. Then $P(B'|A')$ is:

- A. 0
- B. 0.25
- C. 0.5
- D. 1

Quiz: Joint Distributions

X_1 and X_2 have supports $S_1 = \{0, 1\}$ and $S_2 = \{-1, 1\}$, and joint PMF $f(x_1, x_2)$ such that

$$f(0, -1) = \frac{1}{10}, \quad f(0, 1) = \frac{3}{20}, \quad f(1, -1) = \frac{3}{10}, \quad f(1, 1) = \frac{9}{20}.$$

Which claim is correct?

- A. X_1 and X_2 are independent
- B. X_1 and X_2 are not independent
- C. X_1 and X_2 are mutually exclusive
- D. X_1 and X_2 are not mutually exclusive

Review: Continuous Variables

Let X be a random variable with pdf

$$f(x) = \frac{(x+2)}{18}, \quad x \in [-2, 4],$$

Find $P(|X| < 1)$.

Properties of continuous variables

If $a \leq X \leq b$:

- $f(x)$: PDF of X
 - $\int_a^b f(x)dx = 1$
 - $P(X = a) = 0$
- $F(x) = P(X \leq x)$: CDF of X
 - $P(a \leq X \leq b) = \int_b^a f(x)dx = F(b) - F(a)$
 - $P(X \leq a) = P(X < a)$
 - $F'(x) = f(x)$
- $E[X] = \mu_X = \int_a^b xf(x)dx$
- $\text{Var}(X) = \sigma_X^2 = \int_a^b x^2 f(x)dx - \mu_X^2$

Integration by substitution

Let $g(x)$ be a differentiable function in the interval $[a, b]$.

Then for any continuous function $f(x)$ in the interval $[a, b]$

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du,$$

where $u = g(x)$, which implies that $du = g'(x)dx$.

- 1 Find function of x , write as u
- 2 Differentiate u
- 3 Re-calculate integral bounds
- 4 Solve $\int u du$

Practice: u substitution

Compute this integral:

$$\int_0^1 x e^{x^2} dx$$

Practice: Continuous Variables

Let X be a random variable with pdf

$$f(x) = \frac{3(x-2)^2}{35}, \quad x \in [0, c],$$

Find c , $E[X]$, $E[X^2/2]$, and $Var(X)$.

Review: Continuous Variables

Normal Distributions

Normal distribution: classic bell curve, symmetric around mean

$$X \sim N(\mu, \sigma^2), \quad -\infty < x < \infty$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Mean (μ) and variance (σ^2) will always be given to you

Can't integrate $f(x)$ easily

We will almost always convert $N(\mu, \sigma^2)$ to $N(0, 1)$

“Standard” Normal Distributions (Z)

Special case: $\mu = 0$ and $\sigma^2 = 1$; called “standard normal”

$$Z \sim N(0, 1)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x)^2}{2}\right)$$

Use symbol $\Phi(\cdot)$ for CDF of Z

$$F(x) = \Phi(x) = P(Z \leq x)$$

Use a table (like with binomial dist.) for $\Phi(x)$

Standard Normal practice

If $Z \sim N(0, 1)$, find:

- $P(Z \leq 0.87)$
- $P(Z \geq 1.39)$
- $P(0 \leq Z \leq 0.87)$
- $P(|Z| < 1)$
- $P(|Z| > 2)$

Standard Normal rules

If $X \sim N(\mu, \sigma^2)$:

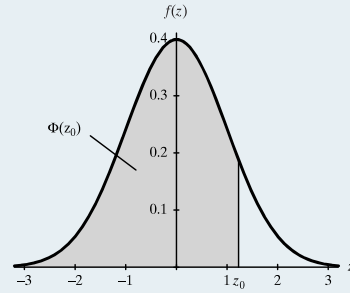
$$\frac{X - \mu}{\sqrt{\sigma^2}} = Z$$

$$\Phi(-x) = 1 - \Phi(x)$$

$$\Phi(0) = 1/2$$

$$z_\alpha \implies \Phi(z_\alpha) = 1 - \alpha \text{ (percentile)}$$

Table Va: The Standard Normal Distribution Function



$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291